

Dimensions

↳ The dimensions of a physical quantity is defined as the powers of the fundamental quantities involved in that physical quantity.

E.g.: The dimensional formula of force is $[MLT^{-2}]$. Hence the dimensions of force are 1 in Mass, 1 in length and -2 in Time.

• Principle of Homogeneity

↳ It states that the dimensions of L.H.S and R.H.S of a correct physical equation must be same.

OR

↳ It states that the dimensions of each term of a correct physical equation must be same.

E.g: In the equation $S = ut + \frac{1}{2}at^2$ we

have dimensions of $S =$ Dimensions of $ut =$ Dimensions of $\frac{1}{2}at^2$

Q.① A student writes an expression of escape velocity as $V = \sqrt{\frac{R}{2GM}}$, where R is radius and M is mass of the earth. Check the correctness of this equation.

↳ Soln,

$$\text{L.H.S} = V$$

$$= [L T^{-1}]$$

$$\text{R.H.S} = \sqrt{\frac{R}{2GM}}$$

$$= \sqrt{\frac{[L]}{[M^{-1} L^3 T^{-2}] [M]}}$$

$$= \sqrt{\frac{[L]}{[L^3 T^{-2}]}}$$

$$= \sqrt{[L^{-2} T^2]}$$

$$= [L^{-1} T]$$

Here,

Dimensions of L.H.S $\neq$ Dimensions of R.H.S

Hence, given equation is dimensionally incorrect.

Q.2 check the correctness of the equation $S = ut + 2at^2$ by dimensional method.

↳ Soln,

$$\begin{aligned} \text{L.H.S} &= S \\ &= [L] \end{aligned}$$

$$\text{R.H.S} = ut + 2at^2$$

$$= [LT^{-1}][T] + 2[LT^{-2}][T^2]$$

$$= [L] + 2[L]$$

$$= 3[L] \quad (\because 3 \text{ is dimensionless quantity})$$

$$= [L]$$

Here,

Dimension of L.H.S = Dimension of R.H.S

Hence, The given equation is dimensionally correct.

Q.3 Is Dimensionally correct equation necessary to be a correct physical equation? What about dimensionally wrong equation?

↳ Dimensionally correct equation is not necessary to be correct physical equation. For e.g.; The equation $K.E = 2mv^2$ is

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dimensionally correct but physically incorrect.

↳ Dimensionally wrong equation is always physically wrong. For e.g.: The equation $F = mv$ is dimensionally as well as physically wrong.

• Application of Dimensional Analysis

- ① To check the correctness of the given equation.
- ② To derive a relation between various physical quantities.

example-1 : Use dimensional method to derive an expression of centripetal force acting on a particle of mass m moving in a circular path of radius r with uniform speed v .

↳ Soln,

It is found that centripetal force acting on a particle depends upon:

① mass of the particle
i.e. $F \propto m^n$

② Radius of the circular path
i.e. $F \propto r^x$

③ Speed of the particle
i.e. $F \propto v^z$

Combining these relations, we get

$$F \propto m^n r^y v^z$$

$$\therefore F = k m^n r^y v^z \rightarrow (i)$$

Where, k is the dimensionless proportionality constant.

Dimensionally, eqn (i) can be written as,

$$[MLT^{-2}] = [M]^n [L]^y [LT^{-1}]^z$$

$$\therefore [MLT^{-2}] = [M^n L^{y+z} T^{-2}]$$

Comparing both sides, we have

$$n = 1$$

$$y + z = 1 \Rightarrow y = 1 - z = 1 - 2 = -1$$

$$-z = -2 \Rightarrow z = 2$$

Putting these values in eqn (i), we get

$$F = k m r^{-1} v^2$$

$$\therefore F = k \frac{m v^2}{r}$$

Since, $k = 1$ (Experimental value)

$$\therefore F = \frac{m v^2}{r}$$

This is required expression of centripetal force.

example - 2: Use dimensional method to derive an expression of time period of a simple pendulum

$$\text{i.e. } T = 2\pi \sqrt{\frac{l}{g}}$$

Where l is the effective length of the pendulum.

→ Soln,

It is found that Time period of a simple pendulum depends upon

① ~~length~~ effective length of pendulum

$$\text{i.e. } T \propto l^n$$

② gravity of the

$$\text{i.e. } T \propto g^y$$

Combining these relations, we get.

$$T \propto l^n g^y$$

$$\therefore T = K l^n g^y \rightarrow (i)$$

Where, K is dimensionless proportionality constant

Dimensionally, eqn (i) can be written as

$$[M^0 L^0 T] = [L]^n [L T^{-2}]^y$$

$$\therefore [M^0 L^0 T] = [L^{n+y} T^{-2y}]$$

Comparing both sides, we have

$$\therefore -2y = 1$$

$$y = -\frac{1}{2}$$

$$x + y = 0$$

$$x = -y$$

$$x = \frac{1}{2}$$

Putting these values in eqn (i) We get,
 $T = k l^{1/2} g^{-1/2}$

$$T = k \frac{l^{1/2}}{g^{1/2}}$$

$$T = k \sqrt{\frac{l}{g}}$$

Since, $k = 2\pi$ (Experimental value)

$$\therefore T = 2\pi \sqrt{\frac{l}{g}}$$

This is required expression of time period of simple pendulum.

③ To convert one system of unit into another.

$$\text{Formula: } N_2 = N_1 \left[\frac{M_1}{M_2} \right]^x \left[\frac{L_1}{L_2} \right]^y \left[\frac{T_1}{T_2} \right]^z$$

Where, x , y and z are the dimensions of mass length and time respectively.

e.g. (i): Convert SN into dyne by dimensional method.

Soln,

S.I system

C.G.S system

$$N_1 = 5$$

$$N_2 = ?$$

$$M_1 = 1 \text{ kg}$$

$$M_2 = 1 \text{ gm}$$

$$L_1 = 1 \text{ m}$$

$$L_2 = 1 \text{ cm}$$

$$T_1 = 1 \text{ s}$$

$$T_2 = 1 \text{ s}$$

Here, Newton is the SI unit of force and dimensional formula of force is $[MLT^{-2}]$

$\therefore$ Dimension of mass (x) = 1

Dimension of length (y) = 1

Dimension of Time (z) = -2

Now,

$$N_2 = N_1 \left[\frac{M_1}{M_2} \right]^x \left[\frac{L_1}{L_2} \right]^y \left[\frac{T_1}{T_2} \right]^z$$

$$= 5 \left[\frac{1 \text{ kg}}{1 \text{ gm}} \right]^1 \times \left[\frac{1 \text{ m}}{1 \text{ cm}} \right]^1 \times \left[\frac{1 \text{ s}}{1 \text{ s}} \right]^{-2}$$

$$= 5 \left[\frac{1000 \text{ gm}}{1 \text{ gm}} \right] \times \left[\frac{100 \text{ cm}}{1 \text{ cm}} \right] \times (1)^{-2}$$

$$= 5 \times 1000 \times 100 \times 1$$

$$= 5 \times 10^5 \text{ dyne}$$

$$\therefore 5 \text{ N} = 5 \times 10^5 \text{ dyne}$$

e.g(ii): Convert 1 erg into Joule by dimensional method.

Soln,

C.G.S system

S.I system

$$N_1 = 1 \text{ erg}$$

$$N_2 = ?$$

$$M_1 = 1 \text{ gm}$$

$$M_2 = 1 \text{ kg}$$

$$L_1 = 1 \text{ cm}$$

$$L_2 = 1 \text{ m}$$

$$T_1 = 1 \text{ sec}$$

$$T_2 = 1 \text{ s}$$

Here, erg is the C.G.S unit of energy and dimensional formula of energy is $[ML^2T^{-2}]$

$\therefore$ Dimension of mass (n) = 1

Dimension of length (y) = 2

Dimension of Time (z) = -2

Now,

$$N_2 = N_1 \left[\frac{M_1}{M_2} \right]^n \left[\frac{L_1}{L_2} \right]^y \left[\frac{T_1}{T_2} \right]^z$$

$$= 1 \left[\frac{1 \text{ gm}}{1 \text{ kg}} \right] \times \left[\frac{1 \text{ cm}}{1 \text{ m}} \right]^2 \times \left[\frac{1 \text{ s}}{1 \text{ s}} \right]^{-2}$$

$$= 1 \left[\frac{1 \text{ gm}}{1000 \text{ gm}} \right] \left[\frac{1 \text{ cm}}{100 \text{ cm}} \right]^2 \times (1)^{-2}$$

$$= 1 \times \frac{1}{1000} \times \frac{1}{10000} \times 1$$

$$= 10^{-7}$$

$$\therefore 1 \text{ erg} = 10^{-7} \text{ J}$$

e.g.-(iii): Convert 13.6 gm/cm^3 into kg/m^3 by dimensional method.

Soln,

C.G.S system

S.I system

$$N_1 = 13.6 \text{ gm/cm}^3$$

$$N_2 = ?$$

$$M_1 = 1 \text{ gm}$$

$$M_2 = 1 \text{ kg}$$

$$L_1 = 1 \text{ cm}$$

$$L_2 = 1 \text{ m}$$

$$T_1 = 1 \text{ s}$$

$$T_2 = 1 \text{ s}$$

Here, gm/cm^3 is the C.G.S unit of density and dimension formula of density is $[ML^{-3}]$

$\therefore$ Dimension of mass (x) = 1

Dimension of length (y) = -3

Now,

$$N_2 = N_1 \left[\frac{M_1}{M_2} \right]^x \left[\frac{L_1}{L_2} \right]^y \left[\frac{T_1}{T_2} \right]^z$$

$$= 13.6 \left[\frac{1 \text{ gm}}{1 \text{ kg}} \right] \times \left[\frac{1 \text{ cm}}{1 \text{ m}} \right]^{-3}$$

$$= 13.6 \left[\frac{1 \text{ gm}}{1000 \text{ gm}} \right] \left[\frac{1 \text{ cm}}{100 \text{ cm}} \right]^{-3}$$

$$= 13.6 \times \frac{1}{1000} \times 1000000$$

$$= 13600$$

$$= 1.36 \times 10^4$$

eg-iv). Find the value of 10 Joule on a system which has 10cm, 100gm and 30 second as the fundamental units.

Soln

S.I System

New system

$$N_1 = 10$$

$$N_2 = ?$$

$$M_1 = 1 \text{ kg}$$

$$M_2 = 100 \text{ gm}$$

$$L_1 = 1 \text{ m}$$

$$L_2 = \cancel{100 \text{ cm}} 10 \text{ cm}$$

$$T_1 = 1 \text{ s}$$

$$T_2 = 30 \text{ sec}$$

Here, Joule is the S.I unit of Work and dimension formula of work is $[ML^2T^{-2}]$

$\therefore$ Dimension of mass (x) = 1

Dimension of length (y) = 2

Dimension of Time (z) = -2

Now,

$$N_2 = N_1 \left[\frac{M_1}{M_2} \right]^x \left[\frac{L_1}{L_2} \right]^y \left[\frac{T_1}{T_2} \right]^z$$

$$= 10 \left[\frac{1 \text{ kg}}{100 \text{ gm}} \right] \times \left[\frac{1 \text{ m}}{\cancel{100 \text{ cm}} 10 \text{ cm}} \right]^2 \times \left[\frac{1 \text{ s}}{30 \text{ s}} \right]^2$$

$$= 10 \times \left[\frac{1000 \text{ gm}}{100 \text{ gm}} \right] \left[\frac{100 \text{ cm}}{10 \text{ cm}} \right]^2 \times \left[\frac{1}{30} \right]^{-2}$$

$$\cancel{= 10 \times 10 \times 100 \times \frac{1}{900}}$$

$$\cancel{\frac{100}{9}} = 10 \times 10 \times 100 \times 900$$

$$= 9 \times 10^6 \text{ new unit}$$

Q To Find the dimensions of constants.

e.g(i). The viscous force acting on a sphere of radius r moving through a viscous medium of coefficient of viscosity η with velocity v is.

$F = 6\pi\eta rv$. Find the dimensions of η .

Soln,

Here, $F = 6\pi\eta rv$

$$\therefore \eta = \frac{F}{6\pi rv}$$

$$= \frac{[MLT^{-2}]}{[L][LT^{-1}]}$$

$$= [ML^{-1}T^{-1}] \quad [\because 6\pi \text{ is dimensionless constant}]$$

$$= [ML^{-1}T^{-1}]$$

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e.g(ii). The displacement of a particle at any instant t is $y = at^2 + bt + c$. Find the dimensions and unit of a, b & c .

Soln,

Here, $y = at^2 + bt + c$

Now,

Dimension of $at^2 =$ Dimension of y

$$\text{or, } a[T^2] = [L]$$

$$\text{or, } a = \frac{[L]}{[T^2]}$$

$$\therefore a = [LT^{-2}]$$

and unit of $a = m/s^2$

Again,

Dimension of $bt =$ dimension of y

$$\text{or, } b [T] = [L]$$

$$\text{or, } b = \frac{[L]}{[T]}$$

$$\therefore b = [LT^{-1}]$$

and Unit of $b = \text{m/s}$

Last,

Dimension of $c =$ dimension of y

$$\text{or, } c = [L]$$

$$\therefore c = [L]$$

and Unit of $c = \text{m}$

e.g (iii). The displacement of a particle at any instant t is $y = a + \frac{b}{c+t^2}$.

Find the dimensions of a, b and c .

Soln,

$$\text{Here, } y = a + \frac{b}{c+t^2}$$

Now,

Dimension of $a =$ Dimension of y

$$\therefore a = [L]$$

$$\text{or, } y = a + \frac{b}{c+t^2}$$

$$\text{or, } y = \frac{a(c+t^2) + b}{(c+t^2)}$$

$$\text{or, } y(c+t^2) = ac + at^2 + b$$

$$\text{or, } yc + yt^2 = ac + at^2 + b$$

Again,

Dimension of $yt^2 =$ Dimension of b

$$\text{or, } b = [L][T^2]$$

$$\text{or, } b = [LT^2]$$

$$\therefore b = [LT^2]$$

Now,

Dimension of $ac =$ Dimension of b

$$\text{or, } [L]c = [LT^2]$$

$$\text{or, } c = \frac{[LT^2]}{[L]}$$

$$[T^2]$$

$$\therefore c = [T^2]$$

e.g - 4 : The displacement of a particle at any instant t is $y = a \sin(\omega t - kx)$ where x is distance. Find the dimensions of a , ω and k .

→ Soln

$$y = a \sin(\omega t - kx)$$

Here, $\sin(\omega t - kx)$ is the dimensionless quantity. So, ~~Dimensions of~~ ~~Time~~

$$\text{Dimension of } a = \text{Dimension of } y \\ = [L]$$

Again, ωt is an angle and is dimensionless quantity.

So,

$$\text{Dimension of } \omega t = [M^0 L^0 T^0]$$

$$\text{or, } \omega [T] = [M^0 L^0 T^0]$$

$$\text{or, } \omega = \frac{[M^0 L^0 T^0]}{[T]}$$

$$\therefore \omega = [M^0 L^0 T^{-1}]$$

Last, $k\pi$ is an angle and is dimensionless quantity.

So,

$$\text{Dimension of } k\pi = [M^0 L^0 T^0]$$

$$\text{or, } k[L] = [M^0 L^0 T^0]$$

$$\text{or, } k = \frac{[M^0 L^0 T^0]}{[L]}$$

$$\therefore k = [M^0 L^{-1} T^0]$$

• Limitation of dimensional analysis.

① The exact correctness of a equation cannot be checked by the dimensional method. Eg: The equation $s = ut + 2at^2$ is dimensionally correct but physically incorrect.

② The equation that contains trigonometric function, exponential function, logarithmic function etc can not be derived by the dimensional method.

Eg: The equation $w = F \cos \theta$ cannot be derived.

③ The equation that contains two or more terms in L.H.S or R.H.S of a equation cannot be derived. Eg: The equation $s = ut + \frac{1}{2}at^2$ cannot be derived.

④ The value of proportionality constant cannot be obtained by the dimensional method. Eg: The value of G cannot be obtained.

⑤ Dimensional method can not be used to identify whether a physical quantity is scalar or vector.

• Precision and Accuracy of measurement

* Precision

↳ The degree of closeness of repeated measurements of a physical quantity is called precision.

If the repeated measurements are less scattered from each other than the measurement is more precise.

* Accuracy

↳ The degree of closeness of observed value of an experiment with its standard value or true value is called accuracy.

If the observed value is less scattered from the standard value then the measurement is more accurate.

Significant figures

↳ The meaningful digits of a number in terms of precision and accuracy ~~is called~~ are called significant figures.

Rules of finding significant figures

① Non-zero digits are always significant.
e.g: 42316 $\longrightarrow$ 5 significant figure.

② Zeros between two non-zero digits are always significant.
e.g: 2003 $\longrightarrow$ 4 significant figures.

③ Leading zeros are never significant.
e.g: 0.002 $\longrightarrow$ 1 significant figure

④ Trailing zeros are significant only if the number contains decimal otherwise they are not significant.
e.g: 2500 $\longrightarrow$ 2 significant figures
2.550 $\longrightarrow$ 4 significant figures

⑤ In the scientific notation $N \times 10^n$, N is significant but 10 and n are not significant.
e.g: $2.50 \times 10^8 \longrightarrow$ 3 significant figures.

Homework

1. The energy of a photon is given by $E = hf$. Find the dimension and unit of Planck's constant h , where f is the frequency of radiation.

→ Given,

$$E = hf$$

$$\text{or, } h = \frac{E}{f}$$

$$\text{or, } h = \frac{[ML^2T^{-2}]}{[T^{-1}]}$$

$$\text{or, } h = [ML^2T^{-2+1}]$$

$$\therefore h = [ML^2T^{-1}]$$

$\therefore$ Dimension of h are 1 in Mass, 2 in length and -1 in Time.

Unit of Planck's constant (h) is $\text{Kgm}^2\text{s}^{-1}$

2. Find the dimension of η in the Stoke's formula, $F = 6\pi\eta r v$, where symbols have their usual meanings.

→ Given,

$$F = 6\pi\eta r v \rightarrow (i)$$

Dimensionally, ~~the~~ eqn (i) can be written as

$$[MLT^{-2}] = \eta [L] [LT^{-1}]$$

($\therefore 6\pi$ is dimensionless quantity)

$$\text{or, } \eta = \frac{[MLT^{-2}]}{[L][LT^{-1}]}$$

$$\text{or, } \eta = \frac{[MLT^{-2}]}{[L^2 T^{-1}]}$$

$$\text{or, } \eta = [ML^{-1} T^{-2+1}]$$

$$\text{or, } \eta = [ML^{-1} T^{-1}]$$

$\therefore$ dimension of η are 1 in Mass, -1 in length and -1 in time.

3. If $y = a + bt + ct^2$, where y is the distance and t is time taken. What is the dimension and unit of c ?

→ Given,

$$y = a + bt + ct^2$$

Here,

Dimension of $ct^2 =$ dimension of y

$$\text{or, } c[T^2] = [L]$$

$$\text{or, } c = \frac{[L]}{[T^2]}$$

$$\text{or, } c = [LT^{-2}]$$

$\therefore$ dimension of c are 1 in length and -2 in time.

$\therefore$ Unit of c is m/s^2 .

4. The force F is given in terms of time ' t ' and displacement ' x ' by the equation $F = A \sin Bx + C \sin Dt$. What is the dimension of D/B .

→ Soln,

Dimension of $Bx =$ Dimension of Dt

$$B[L] = D[T]$$

$$\text{or, } \frac{D}{B} = \frac{[L]}{[T]}$$

$$\text{or, } \frac{D}{B} = [LT^{-1}]$$

$\therefore$ Dimension of $\frac{D}{B}$ are 1 in length and -1 in time.

5. The diameter of a steel rod is given as 56.47 ± 0.02 mm. What does it mean?

→ It means that the diameter of a steel rod is in between 56.45 mm to 56.49 mm and also 0.02 mm is the value of least count of the measuring instrument.

6. Find the dimensions of ~~Planck~~ Planck's constant 'h' from the given equation:

$$\lambda = \frac{h}{P}; \text{ where } \lambda \text{ is wavelength and } P$$

is the momentum of photon.

→ Given,

$$\lambda = \frac{h}{P}$$

$$\text{or, } h = \lambda P$$

$$\text{or, } h = [L][MLT^{-1}]$$

$$\text{or, } h = [ML^2T^{-1}]$$

$\therefore$ Dimensions of Planck's constant 'h' are 1 in mass, 2 in length and -1 in time.

7. Check the correctness of the formula $t = 2\pi \sqrt{\frac{L}{g}}$ using dimensional analysis,

where t is the time period of simple pendulum, L is the length of simple pendulum and g is the acceleration due to gravity.

→ Given,

$$t = 2\pi \sqrt{\frac{L}{g}}$$

L.H.S,

$$t = [T]$$

R.H.S,

$$2\pi \sqrt{\frac{L}{g}} = \sqrt{\frac{[L]}{[L T^{-2}]}} \quad (\because 2\pi \text{ is dimensionless quantity})$$

$$= [L^{1/2}]$$

$$\frac{L^{1/2} T^{-2/2}}{[T^{-1}]}$$

$$= [L^{1/2 - 1/2}]$$

$$= [T^{-1}]$$

$$= \frac{1}{[T^{-1}]}$$

$$= [T]$$

Hence,

$$L.H.S = R.H.S = [T]$$

∴ The given formula is dimensionally correct.

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8. Is dimensionally correct equation necessarily be a correct physical ~~res~~ relation? What about dimensionally wrong equation?

→ No, dimensionally correct equation is not necessarily be a correct physical relation. For example $s = ut + 2at^2$ is dimensionally correct but physically incorrect.

Dimensionally wrong equation are ~~also~~ always physically incorrect. For example $F = mv$ is physically as well as dimensionally incorrect.

9. Convert 10 ergs in Joules using dimensional analysis.

→ Soln,

CGS ~~Unit~~ System

SI System

$$N_1 = 10 \text{ ergs}$$

$$N_2 = ?$$

$$M_1 = 1 \text{ gm}$$

$$M_2 = 1 \text{ kg}$$

$$L_1 = 1 \text{ cm}$$

$$L_2 = 1 \text{ m}$$

$$T_1 = 1 \text{ s}$$

$$T_2 = 1 \text{ s}$$

erg is the CGS unit of work and the dimensional formula of erg is $[ML^2T^{-2}]$

Now,

$$x = 1 \text{ (Dimension of Mass)}$$

$$y = 2 \text{ (Dimension of length)}$$

$$z = -2 \text{ (Dimension of Time)}$$

Using Formula,

$$\begin{aligned} N_2 &= N_1 \left[\frac{M_1}{M_2} \right]^n \left[\frac{L_1}{L_2} \right]^y \left[\frac{T_1}{T_2} \right]^z \\ &= 10 \left[\frac{1 \text{ gm}}{1 \text{ kg}} \right] \left[\frac{1 \text{ cm}}{1 \text{ m}} \right]^2 \left[\frac{1 \text{ s}}{1 \text{ s}} \right]^{-2} \\ &= 10 \left[\frac{1 \text{ gm}}{1000 \text{ gm}} \right] \left[\frac{1 \text{ cm}}{100 \text{ cm}} \right]^2 \times [1]^{-2} \\ &= 10 \times \frac{1}{1000} \times \frac{1}{100} \times \frac{1}{100} \times 1 \\ &= \frac{1}{1000000} \\ &= 10^{-6} \text{ Joule.} \end{aligned}$$

$\therefore 10 \text{ ergs} = 10^{-6} \text{ Joule}$

10. Write any two differences between 'accuracy' and 'precision'.

→ Two differences between 'accuracy' and 'precision' are as follow:

Accuracy	Precision
(i) The degree of closeness of observed value of an experiment with its standard value or true value is called accuracy.	(i) The degree of closeness of repeated measurements of a physical quantity is called precision.
(ii) It indicates the degree of conformity.	(ii) It indicates the degree of reproducibility.

11. The length of rod is exactly 1 cm. An observer records the readings as 1.0 cm, 1.00 cm, and 1.000 cm. Which is the most accurate measurement?

→ 1.000 cm is the most accurate measurement because measurement with more significant figure is more accurate than others.

12. Write the number significant digit in the following:

a) 2.7 = 2 significant figure

b) 30.01 = 4 significant figure

c) 15.0 = 3 significant figure

d) 15.00 = 4 significant figure

e) 0.0 = 1 significant figure.